

UCH401: FLUID AND PARTICLE MECHANICS

L	T	P	Cr
3	1	2	4.5

Course Objective:

To understand basic principles of fluid and particle mechanics including construction and working of the equipment.

Particle Characterization and Handling: Determination of mean particle size, Particle shape and size distribution, Screen analysis, Storage of solids, conveying systems

Size Reduction: Principles of crushing and grinding; Laws of size reduction, Industrial size reduction equipment.

Fluid-Solid Separations: Free and hindered settling, Clarification and thickening, Froth flotation, Centrifugal separation, Theory of filtration and filtration equipment

Packed and Fluidized Bed: Friction in flow through packed beds, Mechanism of fluidization, Determination of minimum fluidization velocity, Determination of velocity range for the operation of a fluidized bed, Applications of fluidized Bed

Agitation and Mixing of Liquids and Solids: Types of impellers, Power consumption, mixing times, Scale up, mixing of solids and power index.

Laboratory Work:

Screen analysis, Power requirement in mixing, Plate and frame filter press, Leaf filter, Elutriation, Pressure drop in fluidized bed and packed bed, Sedimentation, Centrifugal pump characteristics, Size reduction, Cyclone separator.

Course Learning Outcomes (CLOs):

Upon completion of the course, students will be able to:

1. solve and analyze problems of size reduction and solid-solid separation methods.
2. analyze and design of equipment handling fluid-particle systems.
3. analyze and solve problems related to flow through beds of solids.
4. analyze mixing process, and solid storage in hoppers/bins.

Text Books:

1. McCabe, W.L., Smith, J.C., and Harriot, P., *Unit Operations of Chemical Engineering*, McGraw-Hill (2005).
2. Richardson, J.F., Harker, J.H. and Backhurst, J.R., *Coulson and Richardsons Chemical Engineering, Vol. 2*, Butterworth-Heinemann (2007).
3. Rhodes M., *Introduction to particle Technology*, Wiley-Blackwell USA.

Reference Books:

1. Foust, A.S, Wenzel, L.A, Clump, C.W., Maus, L. and Anderson, L.B., *Principles of Unit Operations*, John Wiley (2008).
2. Perry, R.H, and Green, D.W., *Perry's Chemical Engineers' Handbook*, McGraw Hill (2007).
3. Narayanan, C.M. and Bhattacharya, B.C., *Mechanical Operations for Chemical Engineers Incorporating Computer Aided Analysis*, Khanna Publishers (2005).

Evaluation Scheme:

S. No.	Evaluation Elements	Weightage (%)
1	MST	25
2	EST	45
3	Sessional (may include Quizzes/Assignments/Tutorials/Lab evaluations)	30